

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Cross-Platform Mobile Application Development (01CT1517)		Aim: To create an interactive Form using the form widget, Create a Loan Calculator(EMI based on simple as well as compound.	
Experiment No:- 04	Date:- 01-10-2024	Enrollment No:- 92200133030	

Objective:- To create an interactive Form using the form widget, Create a Loan Calculator(EMI based on simple as well as compound.

Code :-

```
import 'package:flutter/material.dart';
import 'dart:math'; // Import math package to use pow() for compound interest calculation

void main() {
  runApp(const MyApp());
}

// Main class for the app
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: ThemeData(
        brightness: Brightness.light,
        primaryColor: Colors.blue, // Light theme with blue as primary color
      ),
      darkTheme: ThemeData(
        brightness: Brightness.dark,
        scaffoldBackgroundColor:
          const Color(0xFF121212), // Dark theme background color
        appBarTheme: const AppBarTheme(
          backgroundColor: Color(0xFF1F1F1F),
          iconTheme: IconThemeData(color: Colors.white),
          titleTextStyle: TextStyle(color: Colors.white, fontSize: 20),
        ),
        textButtonTheme: TextButtonThemeData(
          style: TextButton.styleFrom(
            foregroundColor: Colors.white,
            backgroundColor: Colors.blue,
          ),
        ),
        inputDecorationTheme: InputDecorationTheme(
          filled: true,
          fillColor: Colors.grey[800],
          border: OutlineInputBorder(
            borderRadius: BorderRadius.circular(8.0),
            borderSide: BorderSide.none,
          ),
          hintStyle: const TextStyle(color: Colors.grey),
        ),
        textTheme: const TextTheme(
          bodyMedium: TextStyle(color: Colors.white),
          titleLarge: TextStyle(color: Colors.white),

```

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```

    ),
  ),
  themeMode: ThemeMode.dark, // Set dark mode by default
  home: const EMICalculator(), // Set the EMI calculator screen as the home
  debugShowCheckedModeBanner: false, // Hide the debug banner
);
}
}

// Stateful widget for EMI Calculator
class EMICalculator extends StatefulWidget {
  const EMICalculator({super.key});

  @override
  State<EMICalculator> createState() => _EMICalculatorState();
}

class _EMICalculatorState extends State<EMICalculator> {
  int currentIndex = 0; // 0 for Simple Interest, 1 for Compound Interest
  String result = ""; // To hold the calculated EMI result
  double principal = 0.0; // Principal amount
  double rate = 0.0; // Annual interest rate
  double tenure = 0.0; // Loan tenure in years
  double totalPayment = 0.0; // Total payment calculated
  double totalInterest = 0.0; // Total interest calculated

  // Controllers for TextField inputs
  TextEditingController principalController = TextEditingController();
  TextEditingController rateController = TextEditingController();
  TextEditingController tenureController = TextEditingController();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("EMI Calculator"), // AppBar title
        backgroundColor: Theme.of(context)
          .appBarTheme
          .backgroundColor, // AppBar color based on theme
        actions: [
          IconButton(
            onPressed: () {},
            icon:
              const Icon(Icons.settings), // Settings icon (currently unused)
          ),
        ],
      ),
      body: SingleChildScrollView(
        // Enables scrolling for the screen
        child: Padding(
          padding: const EdgeInsets.all(12.0), // Screen padding
          child: Column(

```

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```

mainAxisAlignment: MainAxisAlignment.start,
crossAxisAlignment: CrossAxisAlignment.start,
children: [
  Row(
    children: [
      // Radio buttons for selecting Simple/Compound interest
      radioButton("Simple Interest", Colors.blue, 0),
      radioButton("Compound Interest", Colors.pink, 1),
    ],
  ),
  const SizedBox(height: 20.0),
  const Text(
    "Loan Amount (Principal) :- ",
    style: TextStyle(fontSize: 18.0),
  ),
  const SizedBox(height: 8.0),
  // TextField for entering principal amount
  TextField(
    controller: principalController,
    keyboardType: TextInputType.number,
    textAlign: TextAlign.center,
    decoration: InputDecoration(
      hintText: "Enter Principal Amount",
      filled: true,
      fillColor: Theme.of(context)
        .inputDecorationTheme
        .fillColor, // TextField styling based on theme
      border: OutlineInputBorder(
        borderRadius: BorderRadius.circular(8.0),
        borderSide: BorderSide.none,
      ),
    ),
  ),
  const SizedBox(height: 20.0),
  const Text(
    "Annual Interest Rate (%) :- ",
    style: TextStyle(fontSize: 18.0),
  ),
  const SizedBox(height: 8.0),
  // TextField for entering interest rate
  TextField(
    controller: rateController,
    keyboardType: TextInputType.number,
    textAlign: TextAlign.center,
    decoration: InputDecoration(
      hintText: "Enter Interest Rate",
      filled: true,
      fillColor: Theme.of(context).inputDecorationTheme.fillColor,
      border: OutlineInputBorder(
        borderRadius: BorderRadius.circular(8.0),
        borderSide: BorderSide.none,
      ),
    ),
  ),

```

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```

    ),
  ),
  const SizedBox(height: 20.0),
  const Text(
    "Loan Tenure (in Years) :- ",
    style: TextStyle(fontSize: 18.0),
  ),
  const SizedBox(height: 8.0),
  // TextField for entering tenure in years
  TextField(
    controller: tenureController,
    keyboardType: TextInputType.number,
    textAlign: TextAlign.center,
    decoration: InputDecoration(
      hintText: "Enter Loan Tenure",
      filled: true,
      fillColor: Theme.of(context).inputDecorationTheme.fillColor,
      border: OutlineInputBorder(
        borderRadius: BorderRadius.circular(8.0),
        borderSide: BorderSide.none,
      ),
    ),
  ),
),
const SizedBox(height: 20.0),
SizedBox(
  width: double.infinity, // Button fills full width
  height: 50.0, // Button height
  child: TextButton(
    onPressed: () {
      // Parsing values from TextField inputs
      setState(() {
        principal =
          double.tryParse(principalController.text) ?? 0.0;
        rate = double.tryParse(rateController.text) ?? 0.0;
        tenure = double.tryParse(tenureController.text) ?? 0.0;
      });
      // Call calculation method based on selected interest type
      if (currentIndex == 0) {
        calculateEMIWithSimpleInterest(principal, rate, tenure);
      } else {
        calculateEMIWithCompoundInterest(principal, rate, tenure);
      }
    },
    child: const Text(
      "Calculate EMI",
      style: TextStyle(color: Colors.white),
    ),
  ),
),
const SizedBox(height: 20.0),
const Center(
  child: Text(

```

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        "Your EMI is :- ",
        textAlign: TextAlign.center,
        style: TextStyle(
          fontSize: 24.0,
          fontWeight: FontWeight.bold,
        ),
      ),
    ),
  ),
  const SizedBox(height: 50.0),
  Center(
    child: Text(
      result, // Display calculated EMI
      textAlign: TextAlign.center,
      style: const TextStyle(
        fontSize: 40.0,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
),
const SizedBox(height: 20.0),
const Center(
  child: Text(
    "Total Payment Breakdown",
    textAlign: TextAlign.center,
    style: TextStyle(
      fontSize: 20.0,
      fontWeight: FontWeight.bold,
    ),
  ),
),
),
const SizedBox(height: 10.0),
// Linear progress bar indicating total payment breakdown
Container(
  height: 20.0,
  decoration: BoxDecoration(
    color: Colors.orange, // Background color of the progress bar
    borderRadius: BorderRadius.circular(10.0),
  ),
),
child: ClipRRect(
  borderRadius: BorderRadius.circular(10.0),
  child: LinearProgressIndicator(
    value: (principal > 0 && totalPayment > 0)
      ? principal / totalPayment
      : 0.0,
    backgroundColor: Colors.transparent,
    valueColor:
      const AlwaysStoppedAnimation<Color>(Colors.green),
  ),
),
),
const SizedBox(height: 10.0),
// Displaying total principal and interest

```


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```

        setState(() {
            currentindex = index; // Update index when radio button is pressed
        });
    },
    style: TextButton.styleFrom(
        backgroundColor: currentindex == index ? color : Colors.grey[200],
    ),
    child: Text(
        value,
        style: TextStyle(
            color: currentindex == index ? Colors.white : color,
        ),
    ),
),
),
),
);
}

```

```

// Method to calculate EMI with Simple Interest
void calculateEMIWithSimpleInterest(
    double principal, double rate, double tenure) {
    setState(() {
        double interest =
            (principal * rate * tenure) / 100; // Simple interest formula
        totalPayment = principal + interest; // Total payment
        totalInterest = interest; // Total interest
        result = (totalPayment / (tenure * 12))
            .toStringAsFixed(2); // EMI calculation for months
    });
}

```

```

// Method to calculate EMI with Compound Interest
void calculateEMIWithCompoundInterest(
    double principal, double rate, double tenure) {
    setState(() {
        double monthlyRate = rate / (12 * 100); // Monthly interest rate
        int n = (tenure * 12).toInt(); // Number of months (n)
        totalPayment =
            principal * pow(1 + monthlyRate, n); // Compound interest formula
        totalInterest = totalPayment - principal; // Total interest
        result =
            (totalPayment / n).toStringAsFixed(2); // EMI calculation for months
    });
}
}

```

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Output:-

EMI Calculator

Simple Interest Compound Interest

Loan Amount (Principal) :-
Enter Principal Amount

Annual Interest Rate (%) :-
Enter Interest Rate

Loan Tenure (in Years) :-
Enter Loan Tenure

Calculate EMI

Your EMI is :-

Total Payment Breakdown

Total Principal ₹ 0.00 Total Interest ₹ 0.00

EMI Calculator

Simple Interest Compound Interest

Loan Amount (Principal) :-
100000

Annual Interest Rate (%) :-
15

Loan Tenure (in Years) :-
5

Calculate EMI

Your EMI is :-

₹ 2916.67

Total Payment Breakdown

Total Principal ₹ 100000.00 Total Interest ₹ 75000.00

EMI Calculator

Simple Interest Compound Interest

Loan Amount (Principal) :-
100000

Annual Interest Rate (%) :-
15

Loan Tenure (in Years) :-
5

Calculate EMI

Your EMI is :-

₹ 3352.26

Total Payment Breakdown

Total Principal ₹ 100000.00 Total Interest ₹ 101135.72